Flask - MySQL

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NEW BUSINESS DIMENSIONS

Cuprins:

MySQL

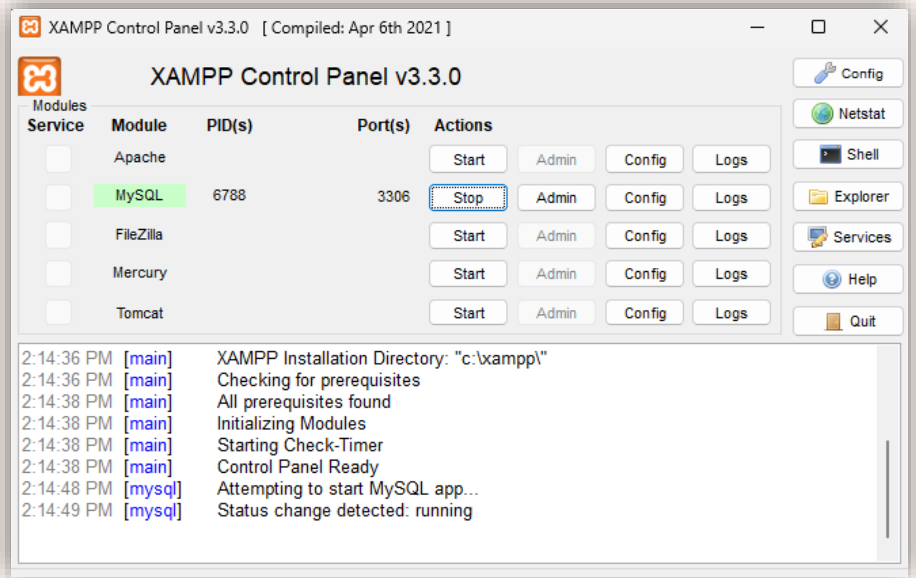
Structured Query Language

SQL este limbajul prin care dai comenzi unei baze de date.

Dacă nu scriem portul, MySQL Connector folosește implicit 3306

```python
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="",
    database="nbd_bank"
)
```

Connector = biblioteca/driverul care face legătura între Python și MySQL.



pip install mysql-connector-python

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.7.0-cp313-cp313-win_amd64.whl.metadata (11 kB)
Downloading mysql_connector_python-9.7.0-cp313-cp313-win_amd64.whl (17.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 17.7/17.7 MB 9.3 MB/s  0:00:01
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.7.0

[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

```python
conn = mysql.connector.connect(
    host="localhost",
    port=3306,
    user="root",
    password="",
    database="nbd_bank"
)
```

C: > Users > Paul > Desktop > f.py > ...

```python
import mysql.connector

# 1. Conectare la MySQL
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password=""
)

cursor = conn.cursor()

# 2. Creare bază de date
cursor.execute("CREATE DATABASE IF NOT EXISTS nbd_bank")

# 3. Selectare bază de date
cursor.execute("USE nbd_bank")

# 4. Creare tabelă
cursor.execute("""
CREATE TABLE IF NOT EXISTS transactions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    client VARCHAR(50),
    amount FLOAT,
    country VARCHAR(10),
    hour INT
)
""")
```

“cursor” - numele vine de la ideea de indicator care se mută prin rezultate.

INT	-- număr întreg
FLOAT	-- număr cu zecimale
VARCHAR	-- text scurt
TEXT	-- text lung
DATE	-- dată
DATETIME	-- dată + oră
BOOLEAN	-- true/false

execută această comandă SQL în MySQL.

“transactions” este numele tabelului.

Creează coloana id.

Creează coloana client. VARCHAR(50) înseamnă text de maximum 50 de caractere.

număr real, cu zecimale.

număr întreg.

```python

# 5. Date de test
data = [
        ("C1", 120, "RO", 14),
        ("C2", 15000, "RU", 2),
        ("C3", 80, "RO", 11),
        ("C4", 9700, "NG", 3),
        ("C5", 300, "FR", 21)
]

# 6. Inserare date
cursor.executemany("""
INSERT INTO transactions (client, amount, country, hour)
VALUES (%s, %s, %s, %s)
""", data)

conn.commit()

```

nbd_bank = baza de date
transactions = tabela
client = coloană
amount = coloană
country = coloană
hour = coloană

executemany() înseamnă: execută INSERT-ul de mai multe ori, câte o dată pentru fiecare rând din var. data.

%s sunt placeholdere, adica locuri rezervate pentru valorile din fiecare rând din data.

```python
# 7. Citire date din tabelă
cursor.execute("SELECT * FROM transactions")

rows = cursor.fetchall()

print("Tranzacții salvate în MySQL:")

for row in rows:
    print(row)

# 8. Închidere conexiune
cursor.close()
conn.close()
```

Daca nu vrem tot (*), putem scrie doar numele coloanelor dorite, ex:
SELECT client, amount FROM transactions

in SQL, * înseamnă tot.

fetchall() = ia toate rândurile returnate de MySQL

RU = Rusia
NG = Nigeria
KP = Coreea de Nord

row = o înregistrare / un rând din tabelă.

cursor = obiect local în Python

```
Tranzacții salvate în MySQL:
(1, 'C1', 120.0, 'RO', 14)
(2, 'C2', 15000.0, 'RU', 2)
(3, 'C3', 80.0, 'RO', 11)
(4, 'C4', 9700.0, 'NG', 3)
(5, 'C5', 300.0, 'FR', 21)
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

Verificari activitati suspecte MySQL

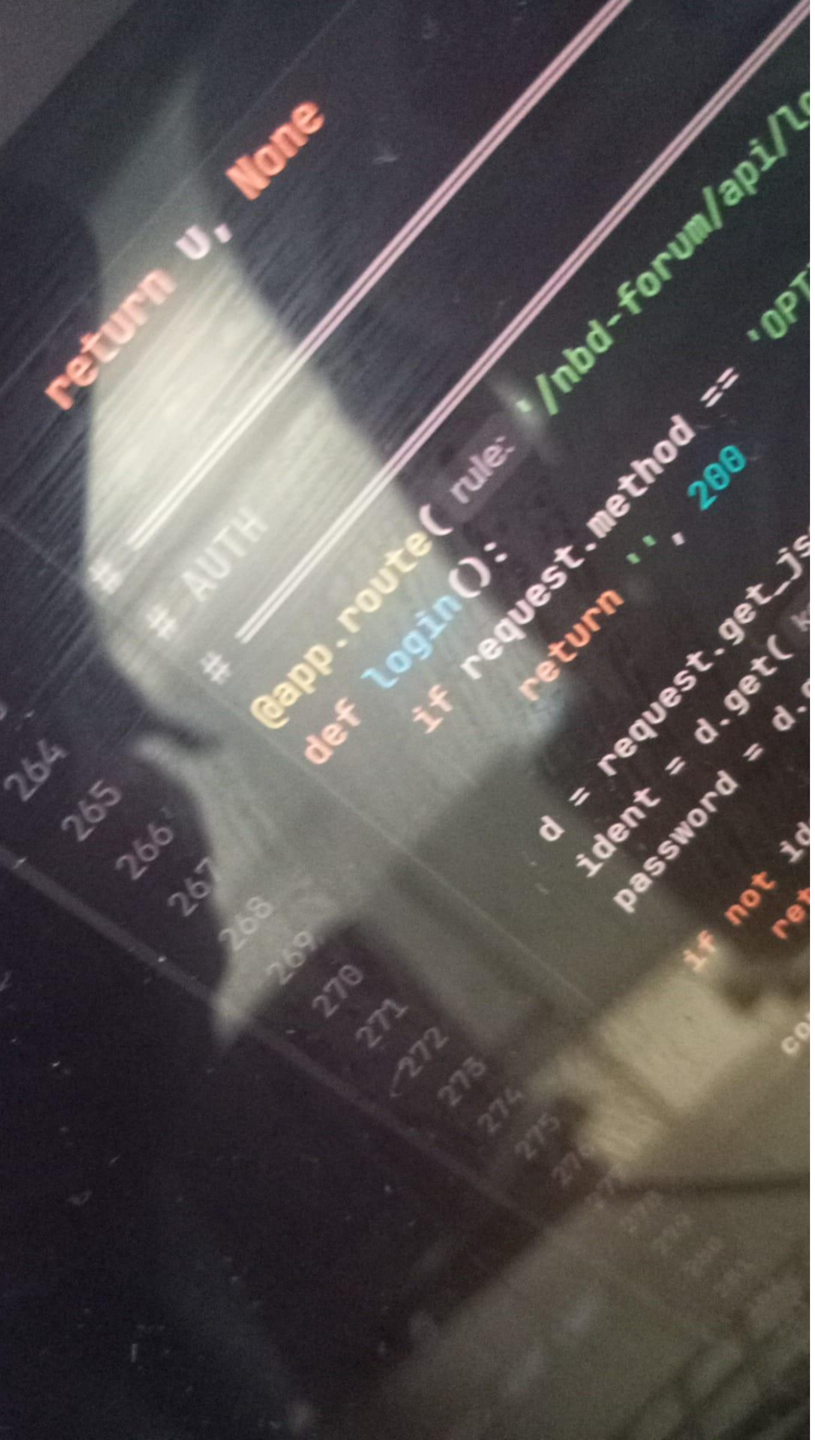
```python
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="",
    database="nbd_bank"
)

cursor = conn.cursor()

query = """
SELECT id, client, amount, country, hour
FROM transactions
WHERE amount > 5000
   OR hour < 5
   OR country IN ('RU', 'NG', 'KP')
"""

cursor.execute(query)

rows = cursor.fetchall()

print("Tranzacții suspecte:")

for row in rows:
    print(row)

cursor.close()
conn.close()
```

Comanda selectează coloanele id, client, amount, country, hour din tabela transactions. Apoi filtrează doar tranzacțiile unde amount > 5000, sau hour < 5, sau country este una dintre valorile RU, NG, KP. Cu alte cuvinte, scoate tranzacțiile considerate suspecte după aceste reguli.

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
Tranzacții suspecte:
(2, 'C2', 15000.0, 'RU', 2)
(4, 'C4', 9700.0, 'NG', 3)
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

RU = Rusia
NG = Nigeria
KP = Coreea de Nord

Flask + MySQL



pip install flask

python -m flask --version

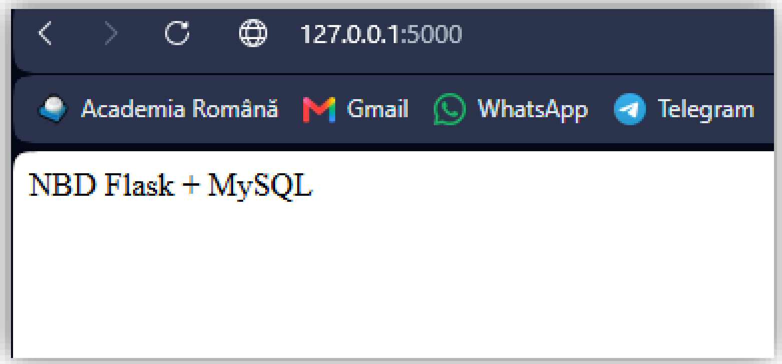
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Requirement already satisfied: colorama in c:\users\paul\appdata\
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.4.1-py3-none-any.whl (116 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading markupsafe-3.0.3-cp313-cp313-win_amd64.whl (15 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
Installing collected packages: markupsafe, itsdangerous, click, b
Successfully installed blinker-1.9.0 click-8.4.1 flask-3.1.3 itsd

[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

Flask instalare

```
[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> python -m flask --version
Python 3.13.7
Flask 3.1.3
Werkzeug 3.1.8
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

Pasul 0 - scheletul minim Flask

```python
from flask import Flask
import mysql.connector

app = Flask(__name__)

def db():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="",
        database="nbd_bank"
    )

@app.route("/")
def home():
    return "NBD Flask + MySQL"

app.run(debug=True)
```



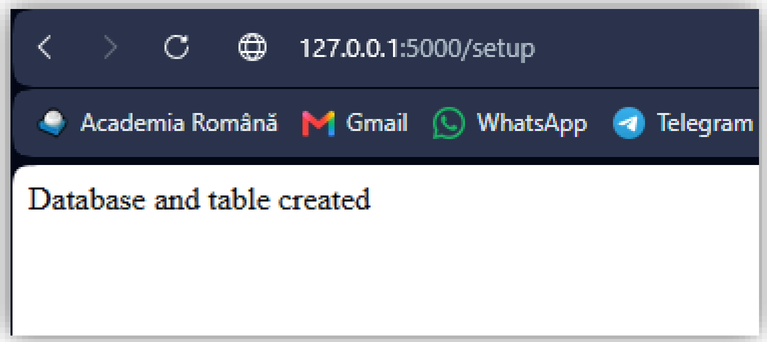

```
PS C:\Users\Paul\AppData\Local\Program
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
```

Pasul 1 - endpoint pentru creare bază și tabelă

Adaugati acest cod înainte de `app.run(debug=True)`:

```python
@app.route("/setup")
def setup():
    conn = mysql.connector.connect(
        host="localhost",
        user="root",
        password=""
    )

    cursor = conn.cursor()

    cursor.execute("CREATE DATABASE IF NOT EXISTS nbd_bank")
    cursor.execute("USE nbd_bank")

    cursor.execute("""
    CREATE TABLE IF NOT EXISTS transactions (
        id INT AUTO_INCREMENT PRIMARY KEY,
        client VARCHAR(50),
        amount FLOAT
    )
    """)

    conn.commit()
    conn.close()

    return "Database and table created"
```

http://127.0.0.1:5000/setup




```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppDa
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 15:55:29] "GET /setup HTTP/1.1" 200 -
```

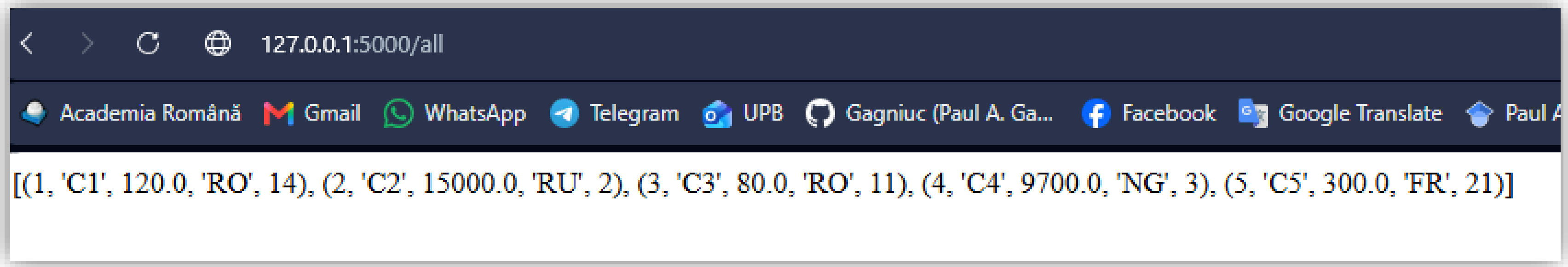
Pasul 2 - endpoint pentru afișarea datelor

Adaugati acest cod înainte de **`app.run(debug=True)`**:

```python

@app.route("/all")
def all_data():
    conn = db()
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM transactions")
    rows = cursor.fetchall()

    conn.close()

    return str(rows)
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a produ
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 16:00:50] "GET /all HTTP/1.1" 200 -
```

http://127.0.0.1:5000/all



La început trebuie să apară: [], dacă tabela este goală. Totuși, în cazul nostru, tabela există deja din exercițiul precedent, deci este posibil să apară înregistrările introduse anterior.

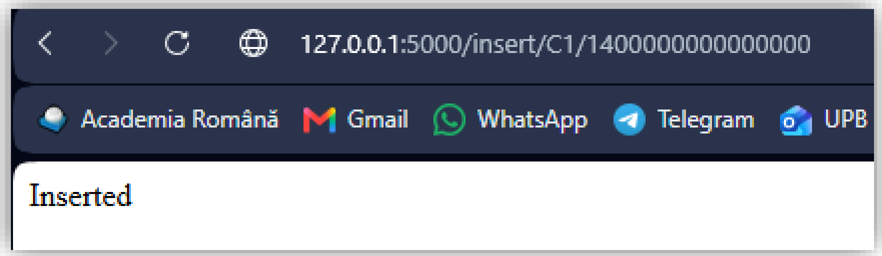
Adaugati acest cod înainte de **`app.run(debug=True)`**:

```python
@app.route("/insert/<client>/<amount>")
def insert(client, amount):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "INSERT INTO transactions(client, amount) VALUES(%s, %s)",
        (client, amount)
    )

    conn.commit()
    conn.close()

    return "Inserted"
```

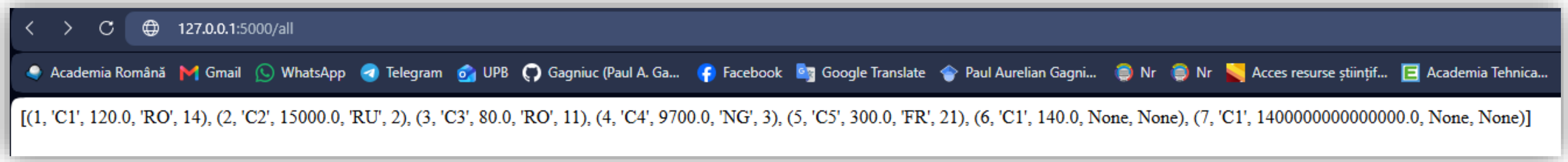
```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Loca
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 16:52:36] "GET /insert/C1/140 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:03] "GET /all HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:04] "GET /all HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:32] "GET /insert/C1/1400000000000000 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 16:53:37] "GET /all HTTP/1.1" 200 -
```

Pasul 3 - endpoint pentru inserare

http://127.0.0.1:5000/insert/C1/14000000000



Deci “country” și “hour” rămân goale.



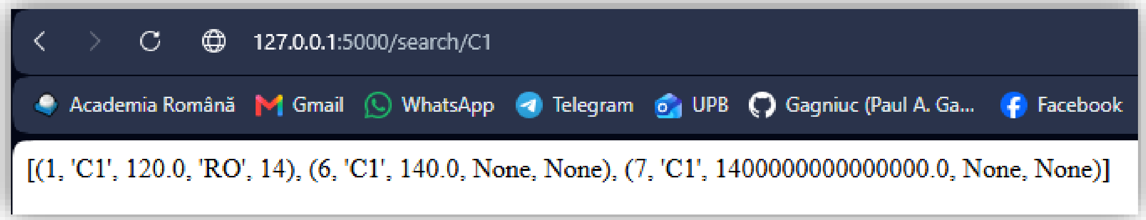
Pasul 4 - endpoint pentru căutare client

Adaugati acest cod înainte de **`app.run(debug=True)`**:

```python
@app.route("/search/<client>")
def search(client):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "SELECT * FROM transactions WHERE client=%s",
        (client,)
    )

    rows = cursor.fetchall()
    conn.close()

    return str(rows)
```

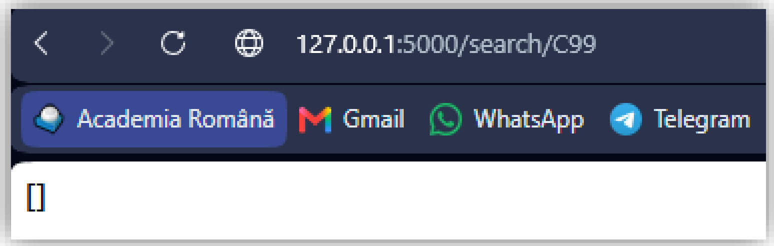
```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppDa
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 17:00:19] "GET /search/C1 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 17:01:34] "GET /search/C99 HTTP/1.1" 200 -
```

http://127.0.0.1:5000/search/C1



Dacă nu există:

http://127.0.0.1:5000/search/C99



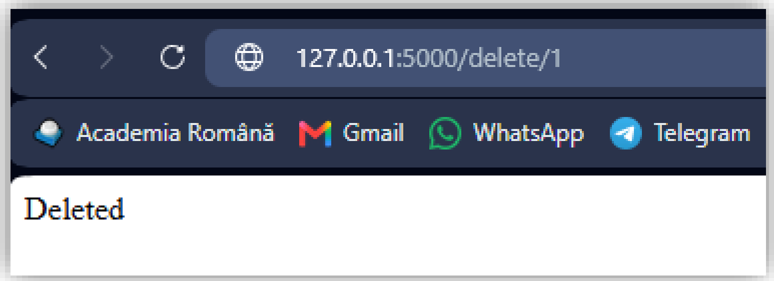
Pasul 5 - endpoint pentru ștergere

Adaugati acest cod înainte de **`app.run(debug=True)`**:

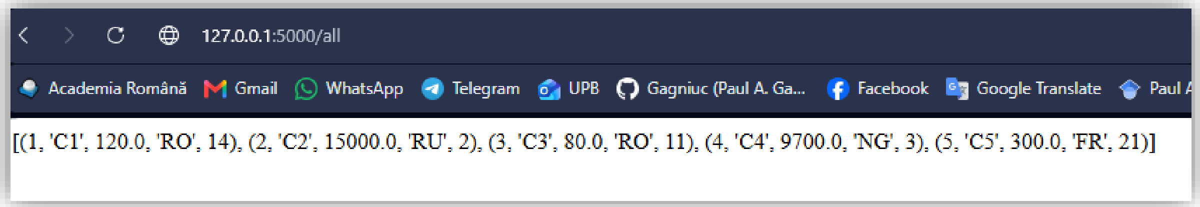
```python
@app.route("/delete/<id>")
def delete(id):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "DELETE FROM transactions WHERE id=%s",
        (id,)
    )

    conn.commit()
    conn.close()

    return "Deleted"
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 18:14:01] "GET /delete/1 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 18:16:03] "GET /all HTTP/1.1" 200 -
```

http://127.0.0.1:5000/delete/1

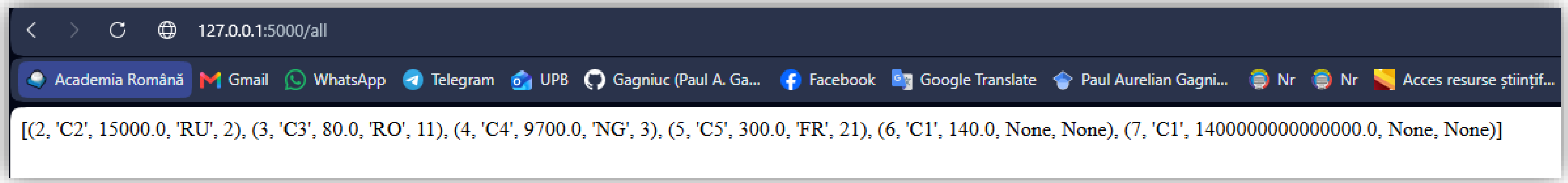


Inainte:
http://127.0.0.1:5000/all



Dupa:
http://127.0.0.1:5000/all

Tranzacția cu id = 1 nu mai apare.



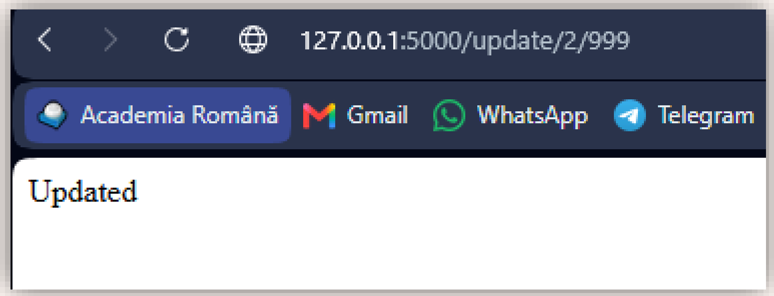
Pasul 6 - endpoint pentru modificare

Adaugati acest cod înainte de **app.run(debug=True)**:

```python

@app.route("/update/<id>/<amount>")
def update(id, amount):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "UPDATE transactions SET amount=%s WHERE id=%s",
        (amount, id)
    )

    conn.commit()
    conn.close()

    return "Updated"

```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppDat
 * Serving Flask app 'f'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 18:24:07] "GET /update/2/999 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 18:28:07] "GET /all HTTP/1.1" 200 -
```

http://127.0.0.1:5000/update/2/999



http://127.0.0.1:5000/all

Trebuie să vezi că suma pentru id = 2 a devenit 999.

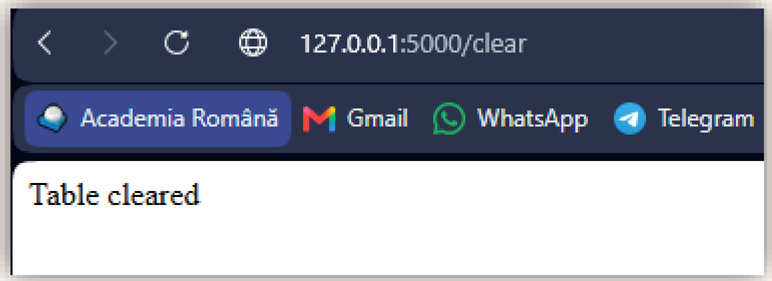
Pasul 7 - ștergere rapidă a tuturor rândurilor

Adaugati acest cod înainte de **`app.run(debug=True)`**:

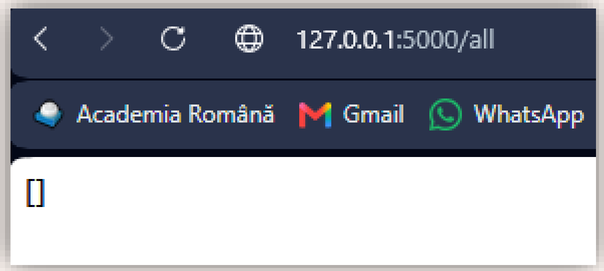
```python

@app.route("/clear")
def clear():
    conn = db()
    cursor = conn.cursor()

    cursor.execute("TRUNCATE TABLE transactions")

    conn.commit()
    conn.close()

    return "Table cleared"

```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users
 * Serving Flask app 'Flask_MySQL'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 19:06:03] "GET /clear HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:07:17] "GET /all HTTP/1.1" 200 -
```

Inainte:
http://127.0.0.1:5000/clear



Dupa:
http://127.0.0.1:5000/clear



TRUNCATE TABLE transactions șterge **toate rândurile** din tabela transactions.

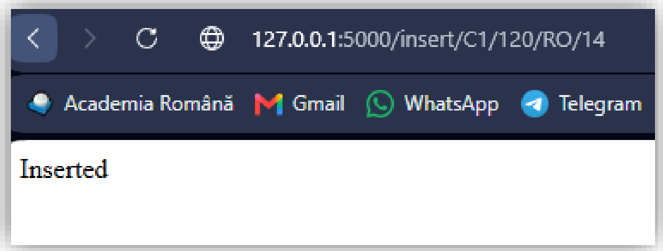
Pasul 8 - endpoint insert corect/complet pentru tabela actuală

Adaugati acest cod înainte de **`app.run(debug=True)`**:

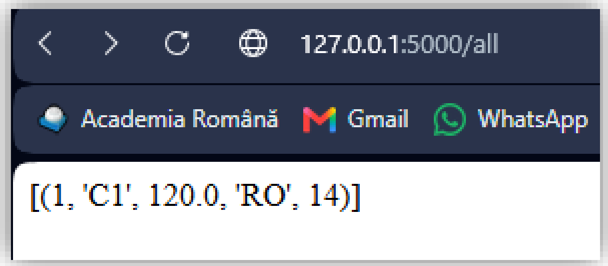
```python
@app.route("/insert/<client>/<amount>/<country>/<hour>")
def insert(client, amount, country, hour):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "INSERT INTO transactions(client, amount, country, hour) VALUES(%s, %s, %s, %s)",
        (client, amount, country, hour)
    )

    conn.commit()
    conn.close()

    return "Inserted"
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppDa
 * Serving Flask app 'Flask_MySQL'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 19:18:14] "GET /insert/C1/120/RO/14 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:19:21] "GET /all HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:20:01] "GET /insert/C1/120/RO/14 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 19:20:04] "GET /all HTTP/1.1" 200 -
```

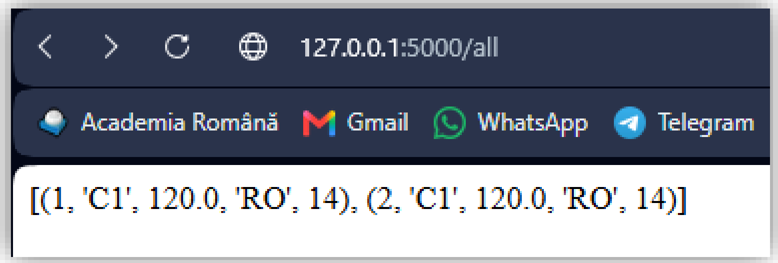
http://127.0.0.1:5000/insert/C1/120/RO/14



http://127.0.0.1:5000/all



2 x http://127.0.0.1:5000/insert/C1/120/RO/14
http://127.0.0.1:5000/all



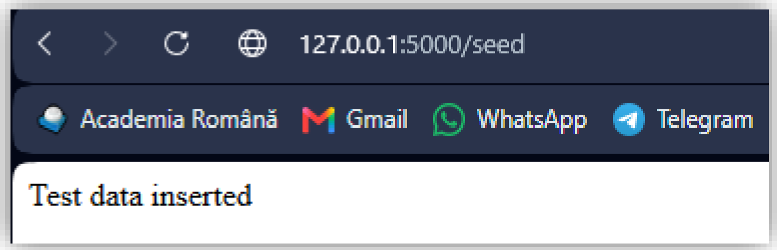
Pasul 9 - încarcă date de test automat

Adaugati acest cod înainte de **`app.run(debug=True)`**:

```python
@app.route("/seed")
def seed():
    conn = db()
    cursor = conn.cursor()

    data = [
        ("C1", 120, "RO", 14),
        ("C2", 15000, "RU", 2),
        ("C3", 80, "RO", 11),
        ("C4", 9700, "NG", 3),
        ("C5", 300, "FR", 21)
    ]

    cursor.executemany(
        "INSERT INTO transactions(client, amount, country, hour) VALUES(%s,%s,%s,%s)",
        data
    )

    conn.commit()
    conn.close()

    return "Test data inserted"
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users
 * Serving Flask app 'Flask_MySQL'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:26:33] "GET /seed HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 21:27:19] "GET /all HTTP/1.1" 200 -
```

http://127.0.0.1:5000/seed



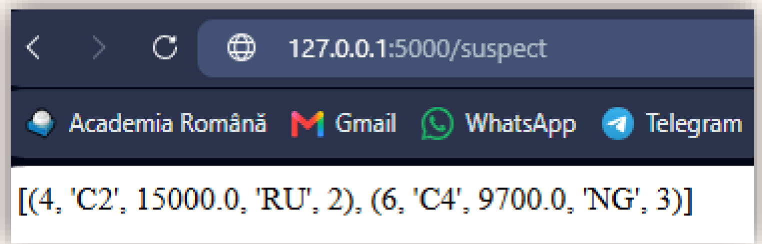
http://127.0.0.1:5000/all

Pasul 10 - tranzacții suspecte

```python


@app.route("/suspect")
def suspect():
    conn = db()
    cursor = conn.cursor()

    cursor.execute("""
        SELECT * FROM transactions
        WHERE amount > 5000 OR hour < 5 OR country IN ('RU', 'NG', 'KP')
    """)

    rows = cursor.fetchall()
    conn.close()

    return str(rows)
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users
 * Serving Flask app 'Flask_MySQL'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:33:07] "GET /suspect HTTP/1.1" 200 -
```

http://127.0.0.1:5000/suspect



Pasul 11 - raport rapid din baza de date

```python
@app.route("/stats")
def stats():
    conn = db()
    cursor = conn.cursor()

    cursor.execute("""
        SELECT
            COUNT(*) AS total_tranzactii,
            SUM(amount) AS suma_totala,
            AVG(amount) AS media
        FROM transactions
    """)

    row = cursor.fetchone()
    conn.close()

    return str(row)
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users
 * Serving Flask app 'Flask_MySQL'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:45:00] "GET /stats HTTP/1.1" 200 -
```

http://127.0.0.1:5000/stats

127.0.0.1:5000/stats

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(7, 25440.0, 3634.285714285714)

COUNT(*) = numără rândurile din tabelă.
SUM(amount) = adună toate sumele.
AVG(amount) = calculează media sumelor.
AS = dă un nume rezultatului.

http://127.0.0.1:5000/all

127.0.0.1:5000/all

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[(1, 'C1', 120.0, 'RO', 14), (2, 'C1', 120.0, 'RO', 14), (3, 'C1', 120.0, 'RO', 14), (4, 'C2', 15000.0, 'RU', 2), (5, 'C3', 80.0, 'RO', 11), (6, 'C4', 9700.0, 'NG', 3), (7, 'C5', 300.0, 'FR', 21)]

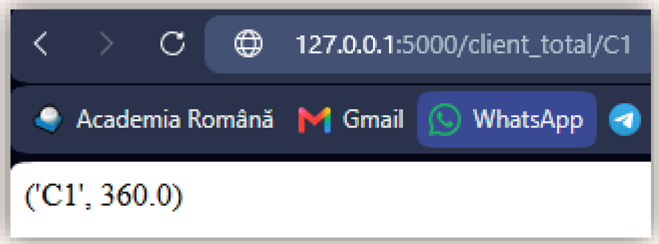
```python
@app.route("/client_total/<client>")
def client_total(client):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "SELECT client, SUM(amount) FROM transactions WHERE client=%s GROUP BY client",
        (client,)
    )

    row = cursor.fetchone()
    conn.close()

    return str(row)
```

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppDat
 * Serving Flask app 'Flask_MySQL'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 456-332-491
127.0.0.1 - - [08/Jul/2026 21:50:53] "GET /client_total/C1 HTTP/1.1" 200 -
127.0.0.1 - - [08/Jul/2026 21:58:02] "GET /client_total/C1 HTTP/1.1" 200 -
```

Pasul 12 - total pe client

Endpointul /client_total/C1 nu afișează o singură tranzacție, ci totalul tuturor tranzacțiilor clientului C1.

http://127.0.0.1:5000/client_total/C1



(1, 'C1', 120.0, 'RO', 14)
(2, 'C1', 120.0, 'RO', 14)
(3, 'C1', 120.0, 'RO', 14)

360

http://127.0.0.1:5000/all

Flask+MySQL+
Criptare

pip install cryptography

```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> pip install cryptography
Collecting cryptography
  Downloading cryptography-49.0.0-cp311-abi3-win_amd64.whl.metadata (4.3 kB)
Requirement already satisfied: cffi>=2.0.0 in c:\users\paul\appdata\local\programs\py
Requirement already satisfied: pycparser in c:\users\paul\appdata\local\programs\pyth
Downloading cryptography-49.0.0-cp311-abi3-win_amd64.whl (3.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 8.3 MB/s  0:00:00
Installing collected packages: cryptography
Successfully installed cryptography-49.0.0

[notice] A new release of pip is available: 25.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

C: > Users > Paul > Desktop > Flask_MySQLcript.py > ...

```python
from cryptography.fernet import Fernet
import base64
import hashlib

def make_key(password):
    return base64.urlsafe_b64encode(
        hashlib.sha256(password.encode()).digest()
    )

def encrypt(text, password):
    f = Fernet(make_key(password))
    return f.encrypt(str(text).encode()).decode()

def decrypt(text, password):
    f = Fernet(make_key(password))
    return f.decrypt(text.encode()).decode()

secret = encrypt("15000", "parola123")
print(secret)

normal = decrypt(secret, "parola123")
print(normal)
```

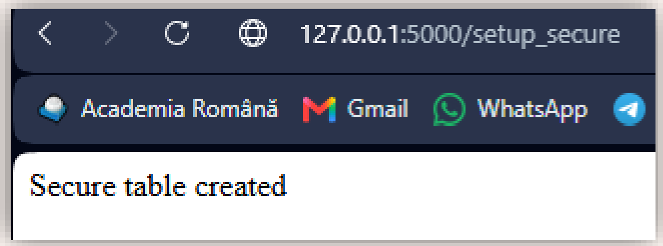
```
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Paul\AppData\Local\Programs\Python
gAAAAABqTrWWP7WBoGxWzYRK4b3om4bsChcUS2UgaO3Fe_YWeFvz_wSa0YhbBS_KBIYuwEQYaGHUwc_Px5HApppI1PwzQ9uuqA==
15000
PS C:\Users\Paul\AppData\Local\Programs\Microsoft VS Code>
```

Inserarea criptată

```python
from flask import Flask
import mysql.connector

from cryptography.fernet import Fernet
import base64
import hashlib

def make_key(password):
    return base64.urlsafe_b64encode(
        hashlib.sha256(password.encode()).digest()
    )

def encrypt(text, password):
    f = Fernet(make_key(password))
    return f.encrypt(str(text).encode()).decode()

def decrypt(text, password):
    f = Fernet(make_key(password))
    return f.decrypt(text.encode()).decode()

app = Flask(__name__)

def db():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="",
        database="nbd_bank"
    )

@app.route("/")
def home():
    return "NBD Flask + MySQL"
```

```python

@app.route("/setup_secure")
def setup_secure():
    conn = db()
    cursor = conn.cursor()

    cursor.execute("DROP TABLE IF EXISTS secure_transactions")

    cursor.execute("""
    CREATE TABLE secure_transactions (
        id INT AUTO_INCREMENT PRIMARY KEY,
        client VARCHAR(50),
        amount TEXT,
        country TEXT,
        hour TEXT
    )
    """)

    conn.commit()
    conn.close()

    return "Secure table created"
```

http://127.0.0.1:5000/setup_secure



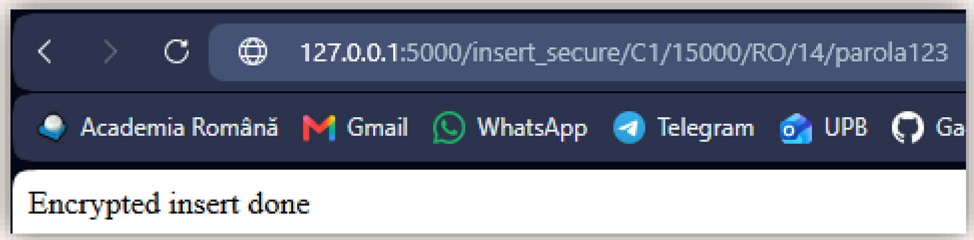
Pentru date criptate, tabela trebuie să fie așa:

id INT AUTO_INCREMENT PRIMARY KEY
client VARCHAR(50)
amount TEXT
country TEXT
hour TEXT

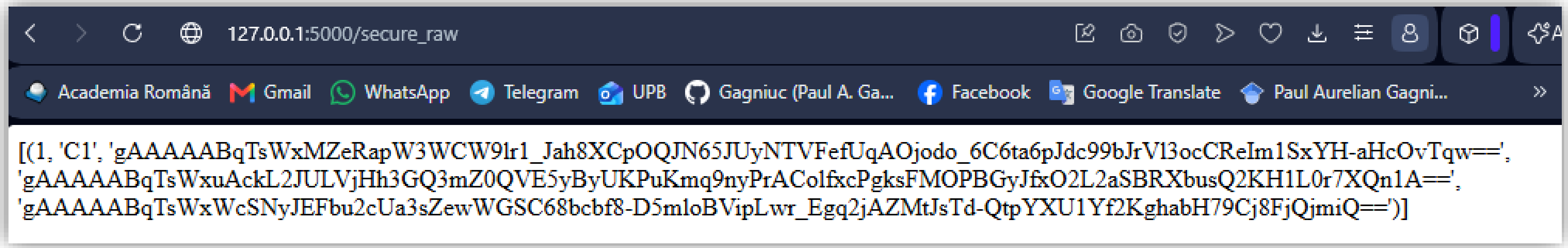
Adică toate coloanele criptate trebuie să fie TEXT.

```python
@app.route("/insert_secure/<client>/<amount>/<country>/<hour>/<password>")
def insert_secure(client, amount, country, hour, password):
    conn = db()
    cursor = conn.cursor()

    encrypted_amount = encrypt(amount, password)
    encrypted_country = encrypt(country, password)
    encrypted_hour = encrypt(hour, password)

    cursor.execute(
        "INSERT INTO secure_transactions(client, amount, country, hour) VALUES(%s, %s, %s, %s)",
        (client, encrypted_amount, encrypted_country, encrypted_hour)
    )

    conn.commit()
    conn.close()

    return "Encrypted insert done"
```

http://127.0.0.1:5000/insert_secure/C1/15000/RO/14/parola123

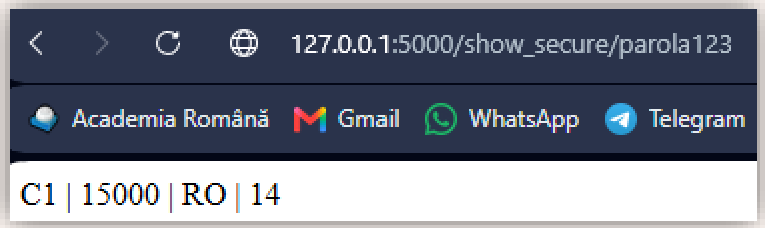


```python
@app.route("/secure_raw")
def secure_raw():
    conn = db()
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM secure_transactions")
    rows = cursor.fetchall()

    conn.close()

    return str(rows)
```



```python
@app.route("/show_secure/<password>")
def show_secure(password):
    conn = db()
    cursor = conn.cursor()

    cursor.execute("SELECT client, amount, country, hour FROM secure_transactions")
    rows = cursor.fetchall()

    result = ""

    for client, amount, country, hour in rows:
        amount_dec = decrypt(amount, password)
        country_dec = decrypt(country, password)
        hour_dec = decrypt(hour, password)

        result += f"{client} | {amount_dec} | {country_dec} | {hour_dec}<br>"

    conn.close()
```

http://127.0.0.1:5000/show_secure/parola123



Cautari approximative.

Ex: LIKE '%ion%' = găsește orice client care conține „ion”

```python
@app.route("/search_like/<name>")
def search_like(name):
    conn = db()
    cursor = conn.cursor()

    cursor.execute(
        "SELECT * FROM transactions WHERE client LIKE %s",
        ("%" + name + "%",)
    )

    rows = cursor.fetchall()
    conn.close()

    return str(rows)

```

http://127.0.0.1:5000/search_like/C


Ză end!